



UNIVERSITÀ
DI TRENTO

High Precision Linear Displacement Measurement System

Precision Engineering

Dipartimento di Ingegneria Industriale

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INDICE

1. System design

- 1.1. Requirements and specifications
- 1.2. Concept generation
- 1.3. Concept selection

2. Detailed design

- 2.1. Analysis
 - 2.1.1. Rigid lever
 - 2.1.2. Optical lever

3. Conclusions

1. System design

The aim of the project consists in creating a High Precision Linear Displacement Measurement System (from now on simply called HP-LDMS).

At the beginning we are aware that we have the maximum freedom of design choices, but we know almost nothing about the problem.

There is no immediate solution at the beginning, so we need to limit the early decisions by maximizing the early problem knowledge.

1.1 Requirements and specifications

In order to do that we establish the list of the customer requirements, so that it is possible to find the corresponding specifications. They are grouped in the House of Quality (QFD):

		Power Consumption Measurement range Temperature range Electronic failure Probe's stiffness # of cycles Weight Set Up Time Dimension Latency Max lateral stiffness Inertia Vibration resistance Electronic noise Variance of the results														
		QFD "Name" Specifications Competitors														
		Variance of the results Electronic noise Vibration resistance Inertia Max lateral stiffness Latency Dimension Set Up Time Weight # of cycles Probe's stiffness Electronic failure Temperature range Measurement range Power Consumption Our Previous Product Master thesis prototype														
		Customers Improvement direction Units nm² dB Hz g/m/s nm s cm³ min kg # N/mm V K mm W/h Ratings														
		Requirements High repeatability Dynamical Properties Easy to maintain Vibration protection Fast response No damage on work piece Good transportability Easy to operate/use High durability Thermally stable Easy to calibrate														
		Competitors Results Our Previous Product Master thesis prototype														
		target (delighted) target (disgusted)														
		1 1 70 / 1 100 30x30 5 1 10⁻⁶ 200 3.3 15 1 5														
		30 20 120 10⁻⁶ 10 300 30x40 30 5 10⁻³ 400 5 5 0.8 10														

The aim of doing the QFD is also to recognise the possible trade off. By finding the best trade offs it will be possible to construct a winning product.

1.2 Concept generation

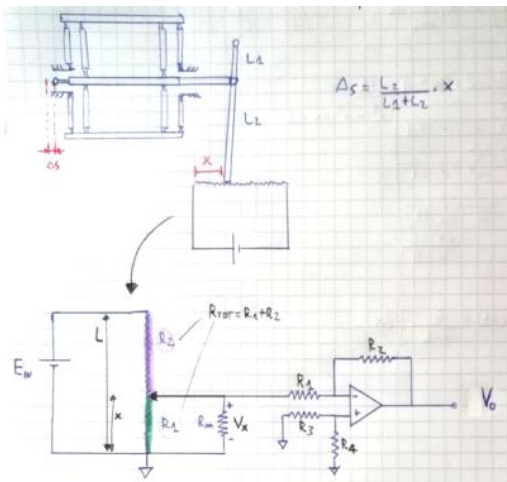
Gradually we move from problem description to the design of the product.

A concept is a description of a physical principle, which idea is based on.

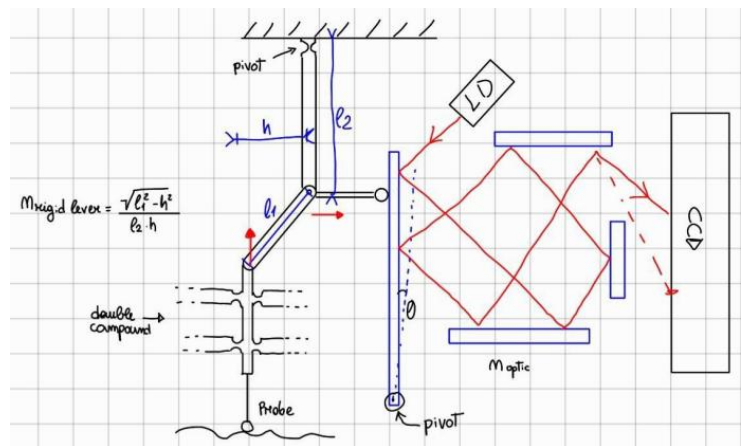
A way to solve the engineering task is to think in parallel many solutions, until I am able to select only one.

Here is shortly presented a list of the main concepts generated:

1) Potentiometer concept



2) Optical mirroring concept



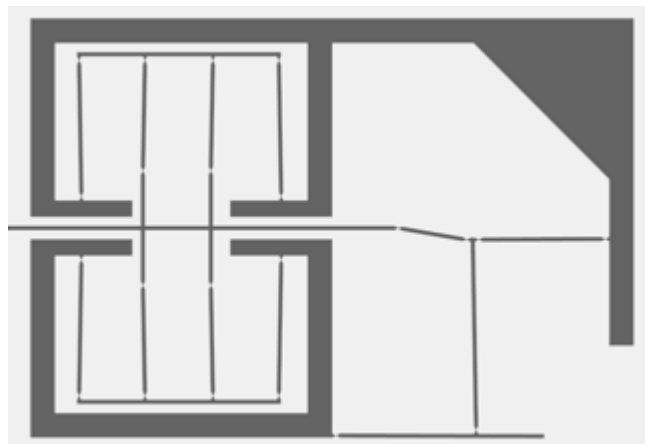
1.3 Concept selection

It is clearly guessable that the potentiometer concept is not really realizable due to the fact that it is needed a high force in input on the probe, to win the frictional force of the pointer. But in order to have high precision and high reliability of the measurement system, the input force must be as low as possible. So, it has been decided to develop the concept that has foundation on the mirroring laser amplification idea.

2. Detailed design

The analysis can be splitted in two parts.

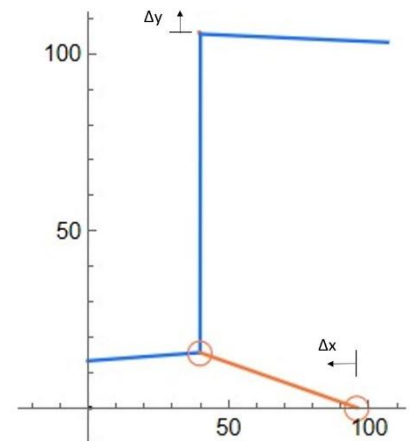
The first part has the aim of finding an input/output relation between the linear displacement of the probe (recognized as input), that is transmitted along the mechanism as the rotation of the mirror (recognized as output). The mirror is fixed with the last beam of the rigid lever, on the bottom side of the figure.



The rigid lever is directly attached to the end of the double compound.

It can be schematized by a four-bar linkage, represented in blue, which has attached a beam with a linear sliding guide, represented in orange.

The input is the linear displacement Δx and the output is a parameter that indicates the rotation of the mirror Δy .



The first step for the realization is to create a model using Simulink's toolbox Simscape Multibody.

The highlighted areas indicate functional modulus:

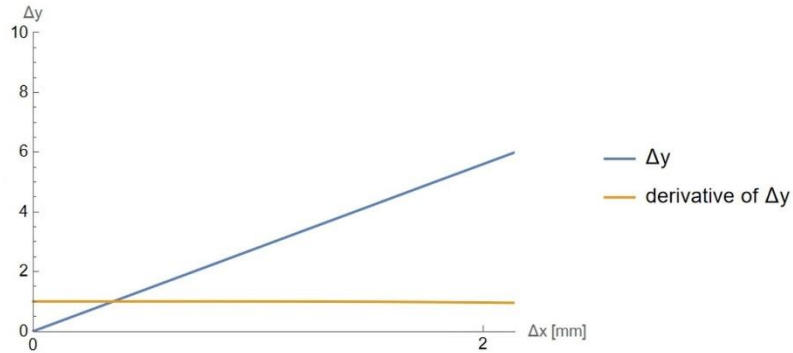
- Double compound (1)
- Rigid lever (2)
- Support structure (3)
- Tilting mirror (4)
- Displacement unit and sensor (5)



From the Matlab analysis the amplification of the rigid lever has been calculated to be nearly 2:

$$\frac{\Delta y}{\Delta x} = \frac{4.121}{2.000} = 2.0605$$

While from the Mathematica simulation it has been possible also to get a graphical representation of the input/output relation:



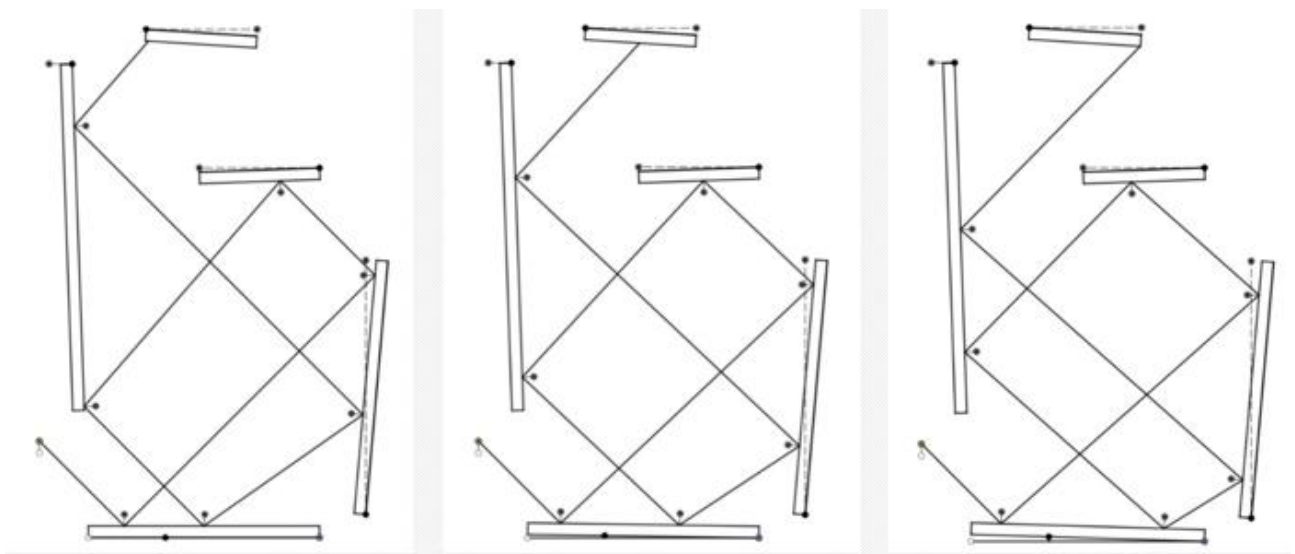
From the derivative curve, that is nearly constant, that for small displacements the behaviour is quasi-linear.

We can also observe that the gain has a value of nearly 2, that concords with the Matlab simulation.

The second part of the analysis regards the optical lever.

The optical lever consists in a total of four mirror, one is the only movable, because it is the one fixed to the mechanism, while the other three are fixed in a particular position that permits to the laser to be reflected several times between them. At the end the laser will hurt a CCD sensor, that has the aim to detect the position of the reflected laser.

Here are presented the possible position of the laser on the CCD sensor, based on the initial angle of the sensor:

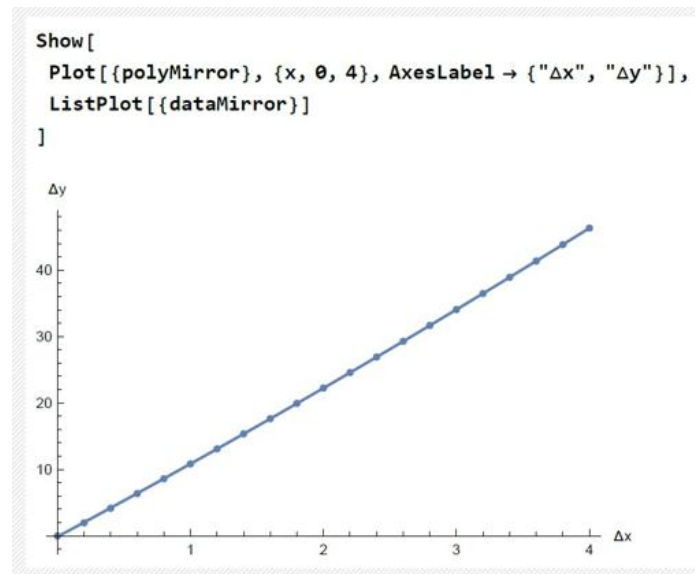


As it is possible to see, the optical lever permits another amplification, that is much bigger than the mechanical one.

Through Fusion 360 we collected some data.

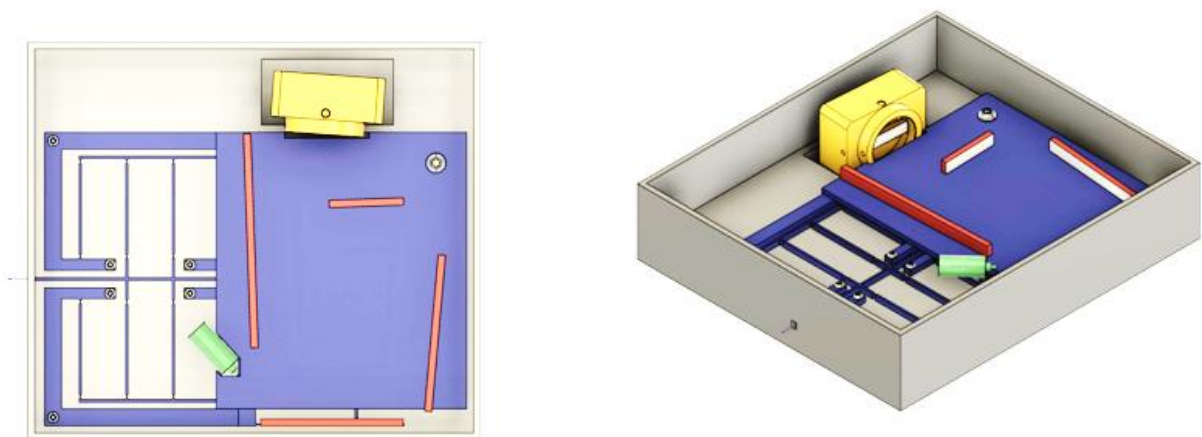
In particular we are interested the output of the CCD sensor and the angle of impact of the laser for a fixed Input displacement.

It has to be noticed that the gain is of about a factor 11. We can see that also from the interpolation graph:



This means, neglecting the non-linearities, that for a displacement range in input of 2mm we can potentially obtain a displacement along the sensor of 22mm.

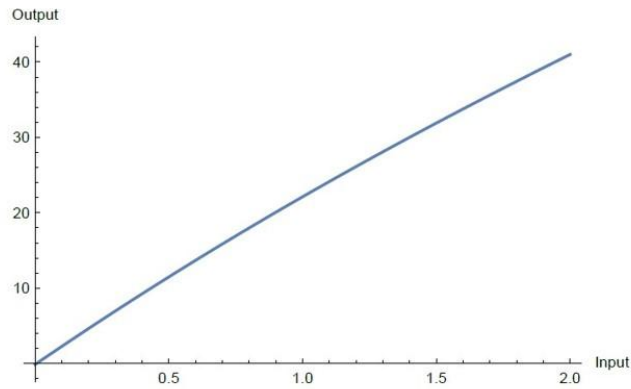
The complete assembly of the measurement system is represented in the following images:



Notice that this is the best assembly possible, because of the space saving by placing the optical lever on the top of the mechanical lever.

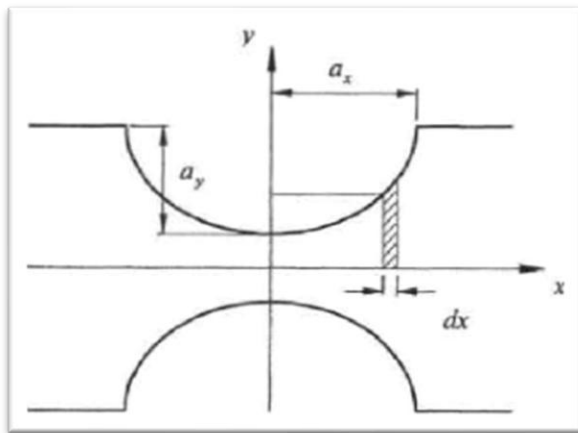
The following figure represent the behaviour of the complete device obtained by passing the rigid lever output as input of the optical lever part:

```
finale = polyMirror /. x -> poly
Plot[{finale}, {y, 0, 2}, AxesLabel -> {"Input", "Output"}]
```



As a final comment on the amplification, we obtained a gain of a factor 40.

In order to have a repeatable system, we opted to use flexure hinges in the double compound and the mechanical lever. Here are reported some parameters and the stiffness.



$$k_{\theta_z, M_z} = \frac{2Ebt^{\frac{5}{2}}}{9\pi a_x^{\frac{1}{2}}} \quad \begin{array}{l} 2a_x = 3mm \\ 2a_y = 1.9mm \end{array}$$

As final element in the following table are reported the information of the CCD sensor, that detect the position of the laser:

Parameter	Specification
CCD	Toshiba TCD1304DG
Number of Pixels	3648
Pixel Size	8μm x 200μm
Spectral Range	300nm to 1050nm 200nm to 1050nm (Windowless version)
Pixel Output Clock	0.5 MHz
Data Storage on Camera	4 Frames
ADC Resolution	16 bits
External Trigger	Yes
Trigger Cable	ACC-CAM-DIN8
Exposure Time Range	0.1ms – 6,500ms
GPIO	Yes (4 programmable I/O's)
Frame Rate	138 scans/second*
Host Interface	USB 2.0

3. Conclusions

We need to obtain a final expression of the whole system performance with related uncertainty on the measured value. In order to do that we performed a Montecarlo analysis splitting the problem in 2 halves: the consideration on the rigid lever and on the optical lever.

As regards the rigid lever we know the analytic relation between input and output, in particular the dimensions that interfere with the system are mainly the oblique bar and the angle that it creates with the horizontal direction. So the precision with which we know the exact measure of the bar and the exact value of the angle characterize the uncertainty of this first step.

We assumed that both measures were distributed as a normal distribution with mean equal to their nominal value and a plausible variance. By applying Montecarlo with $N = 50000$ runs we obtained a first partial result normally distributed. These results will be the input of the next step concerning the optical lever.

Here we have a numerical function but no analytic relation. We determined three parameters (tilting angles of the mirrors, length of the mirrors and thickness of the mirrors) that can influence the final result. By changing them with many simulations we were able to determine the impact of each one of these parameters on the final reading on the CCD sensor surface.

For the tilting angles of mirrors we don't have any introduction error because the mirrors (except the one) are fixed and then for small (less than 0.1°) we don't any change in the result.

The length of the mirrors doesn't influence the final result.

Instead for the thickness we obtain several function depending on each mirror. In particular we can apply the Superposition Principle and we obtain the following results.

This time we considered each uncertainty on the thickness as a noise source, so we considered the thicknesses as normally distributed and we computed the corresponding noises that were all added to the final result obtained by feeding the first step results to the numerical function for the second step. Via Matlab we were able to extract the standard variation of the final distribution and so the uncertainty of the whole system: around 52 nanometers

